Lecture 1: Test lecture

1 Hardy-Weinberg basics

★ Likely tested

L1 deck, slides 2–7

The principle (P1 5)

Genotype frequencies follow $p^2 + 2pq + q^2 = 1$ (P1 5). Allele frequencies hold steady when no force acts on them. (P1 6). The ratio dN/dS compares changes that alter a protein with silent ones. (P1 6).

Write the allele share as f[A1] and the mean fitness as $\bar{w}$ (P1 7). A carrier holds ½ of its alleles in common with a sibling. (P1 7).

Chain: one generation

Entry Parents contribute gametes to one shared pool. (P1 8).

Route Gametes pair at random to form the next generation. (P1 8).

Quick check (P1 9)

Question	Answer	Why
Q1: Is p stable?	Yes, if no force acts	Random mating keeps it fixed (P1 9)
Q2: Sum of p, q?	Exactly 1	Only two alleles exist here (P1 9)

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Where it applies (P2 10)

Small founder groups drift quickly, so their frequencies wander from expectation (P2 10).

Over many generations a population can change until the shift we call evolution has carried it far from where it began.

Reasoning kit

Stated A population is in balance only when five conditions all hold. (P2 11).

Traps

- Do not assume a large population is also randomly mating. (P2 11)
- Do not read a stable p as proof that selection is absent. (P2 12)

Analogy: shuffled deck

Dealing cards from a well shuffled deck never changes how many hearts the deck holds. (P2 12).

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Worked example (P3 13)

Suppose p = 0.6, so q = 0.4 and the heterozygotes make up 2pq = 0.48 of the group. (P3 13). The homozygote classes then hold $p^2 = 0.36$ and $q^2 = 0.16$. (P3 14).

Summing the three classes returns 1, which confirms the bookkeeping. (P3 14).